A Shared LSTM Forecast for 3-D UAV Planning with Coupled MPC–DWA in Dynamic Obstacle Fields

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***Abstract*—First, three-dimensional UAV planning is difficult because moving obstacles can turn, accelerate or pause, making recent positions unreliable. Second, we couple model predictive control (MPC) and the dynamic window approach (DWA) with one causal long short-term memory (LSTM) forecast: MPC optimizes the route, while DWA screens reachable actions and constrains MPC's first command. Then, across 100 simulated scenes with three matched seeds per scene, shared LSTM reached the goal without collision in 73.7% of episodes. This was 16.3 percentage points above a latest-position-hold MPC–DWA reconstruction (simultaneous 95% CI, 5.1–27.6), but below standalone MPC with constant-velocity (CV) prediction (92.0%) and standalone DWA-CV (78.0%). Finally, the results support the shared forecast over the position-hold reconstruction, but not overall superiority across tested controllers.**

***Index Terms*—UAV, 3D path planning, dynamic obstacles, LSTM, model predictive control, dynamic window approach**

I. Introduction

Unmanned aerial vehicles (UAVs) must avoid obstacles that move during flight. Turns, acceleration and pauses make their recent positions unreliable guides to what happens next. Model predictive control (MPC) optimizes a short future route; the dynamic window approach (DWA) chooses a feasible action for the next step [1], [2]. Recent UAV planners also use predicted obstacle motion in MPC [3], [4]. We test whether one causal long short-term memory (LSTM) forecast, shared by MPC and DWA, improves the coupled planner relative to one-layer forecast input, constant-velocity (CV) prediction and standalone controllers. We also compare with a project-local latest-position-hold reconstruction of the MPC–DWA baseline, because the source paper does not specify an executable dynamic-obstacle predictor [5]. The shared forecast connects route optimization to DWA's reachable-action screening and local safety gates.

II. Problem Formulation

We model the UAV as a collision sphere in a bounded three-dimensional workspace with a start, goal and moving spherical obstacles. Each obstacle has an observed center, radius and time-varying path; future ground-truth positions are hidden from the planner. At each update, it chooses a dynamically feasible action to reach the goal without intersecting an obstacle along the simulated motion. Figure 1 shows the scene geometry; recorded planner paths are compared separately.

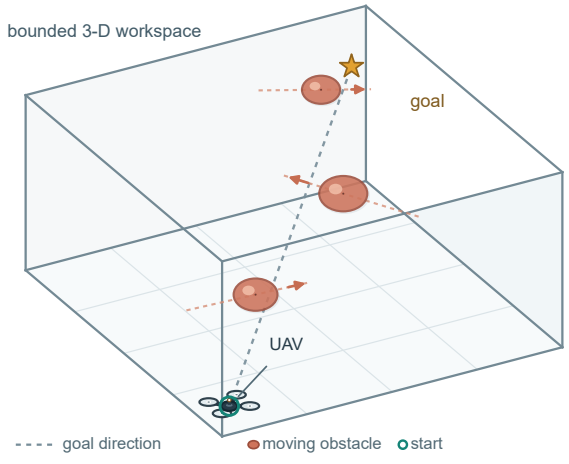


Figure 1. Problem geometry: the quadrotor is centered at the start and moves toward the goal among spherical obstacles following prescribed paths (arrows). The dashed line marks the direct start-to-goal direction, not a simulated route; planner trajectories appear separately in Figure 2.

III. Planning Method

With the task geometry defined, we use one causal obstacle forecast to connect route optimization with locally reachable action selection.

A. Core equations

The vehicle model, causal obstacle forecasts, MPC objective and DWA coupling are collected here. The first four relations update position, speed and attitude; the final relation defines signed surface clearance:

$$\begin{aligned}
\mathbf{p}_{t+1} &= \mathbf{p}_t + \Delta t\, v_t[\cos\theta_t\cos\psi_t, \cos\theta_t\sin\psi_t, \sin\theta_t]^{\mathsf{T}},\\
v_{t+1} &= v_t + \frac{\Delta t}{\tau_v}(v_{c,t} - v_t),\\
\theta_{t+1} &= \theta_t + \Delta t\,\omega_{\theta,t}, \quad \psi_{t+1} = \psi_t + \Delta t\,\omega_{\psi,t},\\
\hat{d}_{j,k} &= \|\mathbf{p}_{t+k|t} - \hat{\mathbf{o}}_{j,t+k|t}\|_2 - r_u - r_j.
\end{aligned} \tag{1}$$

In (1), t and k denote current and future steps; $j = 1, \ldots, M$ indexes the M obstacles. The UAV position, speed, pitch and yaw are $\mathbf{p}$, v, θ and ψ; commands v_c, ω_θ and ω_ψ set speed and angular rates. The update interval is Δt, τ_v is the speed-response time, r_u is the UAV radius and r_j is obstacle j's radius. Thus $\hat{d}_{j,k}$ is signed surface clearance: positive means separated, zero contact and negative overlap.

The constant-velocity (CV) baselines extrapolate from the two latest observed positions; at startup, when only one position is available, the initial velocity is set to zero. A separate latest-position-hold baseline keeps the latest observed center fixed over the forecast horizon. The causal LSTM [6] maps ten consecutive observations to thirty three-dimensional offsets from the latest position:

$$\begin{aligned}
\hat{\mathbf{v}}_{j,t} &= (\mathbf{o}^{\mathrm{obs}}_{j,t} - \mathbf{o}^{\mathrm{obs}}_{j,t-1})/\Delta t,\\
\hat{\mathbf{o}}^{\mathrm{CV}}_{j,t+k|t} &= \mathbf{o}^{\mathrm{obs}}_{j,t} + k\Delta t\,\hat{\mathbf{v}}_{j,t},\\
\mathbf{q}_{j,s} &= [(\mathbf{o}^{\mathrm{obs}}_{j,s})^{\mathsf{T}}, (\mathbf{v}^{\mathrm{obs}}_{j,s})^{\mathsf{T}}, \cos\chi_{j,s}, \sin\chi_{j,s}]^{\mathsf{T}},\\
\hat{\boldsymbol{\delta}}_{j,t+1:t+30|t} &= f_\Theta(\mathbf{q}_{j,t-9:t}),\\
\hat{\mathbf{o}}^{\mathrm{LSTM}}_{j,t+k|t} &= \mathbf{o}^{\mathrm{obs}}_{j,t} + \hat{\boldsymbol{\delta}}_{j,t+k|t}.
\end{aligned} \tag{2}$$

In (2), s indexes history samples; $\mathbf{q}$ contains eight features (3-D position, 3-D velocity, and sine/cosine of heading χ), and f_Θ is the trained LSTM with learned parameters Θ. Here $\mathbf{v}^{\mathrm{obs}}_{j,s}$ is the simulator-provided obstacle velocity, distinct from the finite-difference CV estimate $\hat{\mathbf{v}}_{j,t}$; $\chi_{j,s}$ is obstacle heading (obstacle yaw), not UAV yaw ψ_t. Ten observations span 0.9 s; the LSTM predicts 30 offsets, each anchored to the latest observed center rather than cumulatively summed. At startup, the earliest available state fills missing history. The forecast is causal: at planning time t, the network receives only observations through t. The benchmark compares configured controllers: LSTM receives simulated position, velocity and heading, whereas CV extrapolates the latest two observed positions. Training details are in Supplementary Methods.

In the shared-LSTM arm, one time-aligned forecast is supplied to both planners so MPC's route and DWA's screened actions anticipate the same obstacle motion. This is designed to align long-horizon route optimization with the near-term action check, rather than plan against two different obstacle futures. MPC uses the forecast to optimize its route, while DWA uses corresponding near-term positions to screen reachable commands. Because MPC represents obstacle safety with soft penalties, DWA also enforces explicit clearance, corridor and braking gates. Passing DWA actions define the componentwise box that constrains MPC's first command in (3). In the historical DWA-only ablation, LSTM informs DWA while MPC holds the latest observed obstacle positions. The MPC objective and coupling constraints are:

$$\begin{aligned}
\mathbf{a} &= \mathbf{g} - \mathbf{s},\\
\eta_{\mathbf{a}} &= \mathbf{a}^{\mathsf{T}}\mathbf{a} + 10^{-12}\,\mathrm{m}^2,\\
P_{\mathbf{a}} &= I_3 - \frac{\mathbf{a}\mathbf{a}^{\mathsf{T}}}{\eta_{\mathbf{a}}},\\
\mathbf{e}^{\perp}_{t+k|t} &= P_{\mathbf{a}}(\mathbf{p}_{t+k|t} - \mathbf{s}),\\
J_r &= \sum_{k=1}^{30} \|\mathbf{e}^{\perp}_{t+k|t}\|_2^2,\\
\min_{\mathbf{u}_1,\ldots,\mathbf{u}_{30}} \; J_{\mathrm{MPC}} &= \sum_{k=0}^{29} w_p\|\mathbf{p}_{t+k|t} - \mathbf{g}\|_2^2\\
&\quad + \sum_{k=1}^{30} w_u\|\mathbf{u}_k\|_2^2\\
&\quad + w_f\|\mathbf{p}_{t+30|t} - \mathbf{g}\|_2^2 + w_r J_r\\
&\quad + w_{\Delta u}\sum_{k=2}^{30}\|\mathbf{u}_k - \mathbf{u}_{k-1}\|_2^2\\
&\quad + \sum_{k=1}^{30}\sum_{j=1}^{M} w_o[-\hat{d}_{j,k}]_+^2\\
&\quad + \sum_{k=1}^{30}\sum_{j=1}^{M} w_s[d_{\mathrm{soft}} - \hat{d}_{j,k}]_+^2,\\
&\text{subject to } \mathbf{x}_{t|t} = \mathbf{x}_t,\\
&\mathbf{x}_{t+k|t} = f(\mathbf{x}_{t+k-1|t}, \mathbf{u}_k),\\
&\qquad k = 1, \ldots, 30,\\
&\mathbf{u}_k \in [0, 2.0] \times [-0.8, 0.8]^2,\\
&\mathbf{u}_1 \in \mathcal{U}_t \quad \text{when } \mathcal{A}_t \neq \varnothing,\\
&|\mathbf{u}_k - \mathbf{u}_{k-1}| \preceq \mathbf{a}_u\Delta t,\\
&\qquad k = 1, \ldots, 30.
\end{aligned} \tag{3}$$

Here $\mathbf{x}_t = [\mathbf{p}_t^{\mathsf{T}}, \theta_t, \psi_t, v_t]^{\mathsf{T}}$ is the position–pitch–yaw–speed state; f is defined by the first four state-update relations in (1). The MPC command is $\mathbf{u}_k = [v_{c,k}, \omega_{\theta,k}, \omega_{\psi,k}]^{\mathsf{T}}$; $\mathbf{u}_0$ is the last applied DWA command and $\mathbf{a}_u$ gives componentwise rate limits. The physical input box means commanded speed is 0–2.0 m s^{-1} and each angular rate is limited to ± 0.8 rad s^{-1}. In the rate constraint, absolute value and $\preceq$ act componentwise, so each command component may change by at most its corresponding limit in $\mathbf{a}_u$ over Δt. The start and goal are $\mathbf{s}, \mathbf{g}$, and $\mathbf{a} = \mathbf{g} - \mathbf{s}$ is the route vector. The scalar $\eta_{\mathbf{a}}$ is its squared length plus the implementation's $10^{-12}\,\mathrm{m}^2$ numerical safeguard; $P_{\mathbf{a}}$ is the resulting regularized cross-track operator,

and J_r penalizes the residual distance from the direct line. I_3 is the three-dimensional identity. The index k runs over the prediction horizon and $\hat{d}_{j,k}$ is signed clearance for obstacle j; MPC adds a $10^{-12}\,\mathrm{m}^2$ numerical smoothing term when evaluating distance, whereas DWA and swept-collision checks use the exact norm. Weights $w_p, w_u, w_f, w_r, w_{\Delta u}, w_o, w_s$ penalize, respectively, stage goal distance, control effort, terminal distance, route deviation, command changes, overlap and clearance below $d_{\mathrm{soft}} = 0.35$ m. The positive-part operator is $[x]_+ = \max(0, x)$. The implementation squares native command values without cross-channel normalization, making the effort term a heuristic scalarization rather than a dimensionally homogeneous physical cost; numerical weights and command limits are given in Supplementary Methods. The MPC solver imposes no separate state-box bounds. When the normal safe set $\mathcal{A}_t$ is nonempty, its passing actions provide componentwise bounds $\mathcal{U}_t$ for MPC's first input; this box constrains MPC but is not itself a safety certificate for every enclosed action.

Trust score τ combines predicted safety, goal progress, agreement with the previous MPC guide, and control smoothness. It sets the sample budget and guided/exploratory split. Components and weights lie in $[0, 1]$; weights sum to one.

$$\begin{aligned}
\tau &= \sigma_s \tau_{\mathrm{sa}} + \sigma_g \tau_g \\
&\quad + \sigma_t \tau_{\mathrm{ref}} + \sigma_m \tau_{\mathrm{sm}}, \\
\Delta_p &= \|\mathbf{p}_t - \mathbf{p}_{\mathrm{ref},1|t-1}\|_2, \\
\tau_{\mathrm{ref}} &= \exp(-\alpha_T \Delta_p), \\
B &= \mathrm{round}[300(1 + \alpha e^{-\beta\tau})], \\
B_g &= \mathrm{round}(B\tau), \quad B_e = B - B_g.
\end{aligned} \tag{4}$$

In (4), $\sigma_s, \sigma_g, \sigma_t, \sigma_m$ weight safety, goal progress, reference tracking and smoothness; they sum to one, and σ_t denotes tracking rather than time. The reference position $\mathbf{p}_{\mathrm{ref},1|t-1}$ is the first predicted point on the previous MPC route, so Δ_p measures the UAV's distance from that guide. By definition, τ_{ref} decreases with this distance; the implementation names this tracking score τ_t. At startup, scores use the initial MPC solution; later they use the shifted route and controls. The four scores measure clearance-based safety, route progress, agreement with the previous route's first point, and control smoothness; their full definitions are in Supplementary Methods S1.2. The historical benchmark retained inherited weights $[0.35, 0.30, 0.15, 0.20]$, not tuned on its 500 scenes. A separate 4,320-episode development sweep (six vectors, five integrated branches, 48 scenes and three matched seeds) selected $[0.15, 0.15, 0.15, 0.55]$, with equal-arm macro-average collision-free arrival of 0.8042 (branch rates in Supplementary Methods S1.2). Because the same runs selected the vector, this score is descriptive, not held-out evidence: the historical benchmark retained its inherited vector, while the current 100-scene evaluation uses the selected vector. Its $\alpha_T = 3\ \mathrm{m}^{-1}$ and dimensionless $\beta = 3$ settings remained fixed and were not selected by the incomplete Stage-2 grid. B is split into MPC-guided (B_g) and exploratory (B_e) samples, and the clipped MPC command is also tested. The dimensionless budget gain $\alpha = 1$ is distinct from α_T; higher trust reduces exploration and sample count, 300 is the base budget, and β sets sensitivity.

DWA rejects candidate b unless its rollout meets the clearance, flight-corridor and braking checks. It ranks the remaining actions and bounds the MPC first control using their componentwise control limits:

$$\begin{aligned}
\mathcal{A}_t &= \{b : d_{\min}^{(b)} > 0.25\ \mathrm{m},\ \mathcal{C}_b = 1, \\
&\qquad (v_{\mathrm{end}}^{(b)})^2 \le 2 a_{\max} d_{\min}^{(b)}\}, \\
S_b &= w_p^{\mathrm{DWA}} S_{p,b} + w_h^{\mathrm{DWA}} S_{h,b} \\
&\quad + w_v^{\mathrm{DWA}} S_{v,b} + w_o^{\mathrm{DWA}} S_{o,b} + w_z S_{z,b}, \\
b^* &= \arg\max_{b \in \mathcal{A}_t} S_b, \\
u_{i,t}^- &= \max(u_i^{\min}, \min_{b \in \mathcal{A}_t} u_i^{(b)} - \delta), \\
u_{i,t}^+ &= \min(u_i^{\max}, \max_{b \in \mathcal{A}_t} u_i^{(b)} + \delta), \\
\mathcal{U}_t &= \prod_{i=1}^{3} [u_{i,t}^-, u_{i,t}^+].
\end{aligned} \tag{5}$$

In (5), candidate b passes when its discrete predicted clearance $d_{\min}^{(b)} > 0.25$ m, corridor flag $\mathcal{C}_b = 1$ and braking test pass; $v_{\mathrm{end}}^{(b)}$ and $a_{\max}$ are terminal speed and available deceleration. Scores S_p, S_h, S_v, S_o measure MPC-endpoint agreement, heading agreement, speed tracking with a forward-speed reward, and normalized obstacle clearance; S_z penalizes vertical-profile deviation. Trust-dependent weights on the first four scores sum to one; $w_z = 0.25$ is fixed and not renormalized. The margin $\delta = 0.1$ is applied separately to each command coordinate (not as one shared physical distance) to expand the screened candidates' range; i indexes speed, pitch rate and yaw rate, and $u_i^{\min}, u_i^{\max}$ are physical limits. The top-scoring safe candidate is normally applied. A yaw-rate correction is triggered when the selected rollout's minimum predicted clearance exceeds 1.0 m while the UAV's current horizontal cross-track offset from the start–goal line exceeds 0.5 m; the adjusted command is re-screened by the same clearance, corridor and braking tests, and the original safe command is restored if it fails. Collision outcomes are independently checked on continuous swept paths. If the normal set is empty, the emergency branch is screened and the run stops if none passes. Full score formulas and settings are in Supplementary Methods.

B. How the shared forecast is used

At each 0.1-s update, the shared arm sends the same causal 3-s LSTM forecast to MPC and DWA. MPC optimizes a 30-step route; DWA screens reachable 30-step actions against that forecast, and its passing actions bound MPC's first command on re-solve. The updated MPC route guides the next DWA ranking, while the screened DWA action is executed. In the one-layer ablations, only the named layer receives LSTM predictions and the other uses CV; independent DWA and MPC use CV, and the no-forecast comparator holds the latest obstacle positions. The complete cycle and fallback are in Supplementary Methods S1.4.

IV. Simulation and Results

With the controller roles defined, we compared seven arms in a limited-scope exploratory simulation. A new bank of 100 scenes was evaluated with three matched sampling seeds per scene, giving 300 episodes per arm. All controllers updated every 0.1 s and used 30-step (3-s) MPC and DWA horizons, with a 650-step limit. The arms were standalone DWA-CV, standalone MPC-CV, coupled MPC–DWA-CV, LSTM to MPC only, LSTM to DWA only, shared LSTM, and a project-local latest-position-hold MPC–DWA reconstruction. In each one-layer arm, the other layer used CV; the shared arm sent the same causal LSTM forecast to both layers. The primary outcome was arrival within goal tolerance without contact on the continuous swept path. We averaged seeds within each scene and used 20,000 scene-cluster bootstrap resamples; simultaneous max-—t— intervals covered the six shared-versus-comparator contrasts. This new exploratory comparison is reported separately from the historical five-controller study in the Supplement.

Shared LSTM reached the goal without collision in 221/300 episodes (73.7%). Its paired advantage over the position-hold reconstruction was 16.3 percentage points (simultaneous 95% CI, 5.1–27.6). It was lower than standalone MPC-CV by 18.3 points (-29.6 to -7.0); intervals for standalone DWA-CV, coupled MPC–DWA-CV and both one-layer LSTM arms included zero. Thus, the data do not establish that sharing is best among the tested controllers.

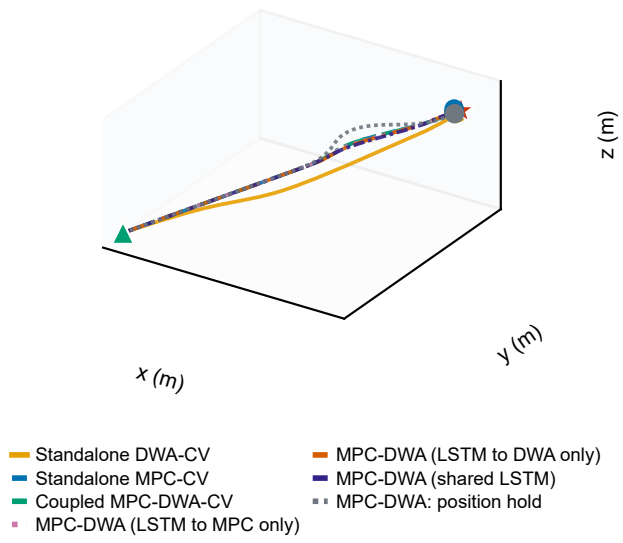


Figure 2. Recorded three-dimensional paths for all seven arms in the preselected matched scene (scene 1, seed 1). Each path ends at its recorded terminal state; none is smoothed, resampled or extended. The triangle and star mark the shared start and goal; circles mark collision-free goal arrivals, squares mark no-contact non-arrivals, and X marks a terminal state from a run with swept contact, not necessarily the contact location. Colors and line styles identify the arms. Axes are in metres; numeric ticks and obstacle tracks are omitted from this route overlay, while prescribed obstacle motion is shown in Fig. 1. All seven arms reached the goal without contact in this example. It illustrates route geometry only; Table 1 reports aggregate outcomes.

Table 1
Outcome summary for the exploratory simulation ($n = 300$ episodes per arm). Safe arrival requires reaching the goal without swept-path contact. Full paired intervals are in Supplementary Table S12.

Controller	Safe-goal rate (%)	Swept contacts
Standalone DWA-CV	78.0	0
Standalone MPC-CV	92.0	15
Coupled MPC–DWA-CV	68.0	0
MPC–DWA (LSTM to MPC only)	73.0	0
MPC–DWA (LSTM to DWA only)	74.7	0
MPC–DWA (shared LSTM)	**73.7**	**0**
MPC–DWA (latest-position hold)	57.3	50

V. Discussion

Sharing the forecast improved arrival over the position-hold reconstruction, but its estimate did not exceed the coupled CV arm or either one-layer LSTM arm. Standalone MPC-CV had higher safe-arrival frequency, with its paired interval favoring that baseline; standalone DWA-CV was numerically higher, but its simultaneous interval included zero. These results favor a limited conclusion about one forecast-allocation contrast, not an overall ranking. The benchmark uses prescribed simulated tracks and a point forecast, and includes no real-flight validation.

VI. Conclusion

In this exploratory simulation, the shared causal LSTM forecast improved safe arrival over a latest-position-hold MPC–DWA reconstruction, while standalone MPC-CV performed better and the remaining paired comparisons were inconclusive. The results motivate further evaluation of shared prediction in dynamic-obstacle planning but do not establish overall superiority or real-flight performance.

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